

Solving coordination games with minimal communication

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1 Introduction

In our social lives, we regularly face group decision problems: what should *we* do? Each of us must select an individual action in such a way that their combination gives rise to a collective action that solves the problem. Collective actions, at least ontically, are nothing but agglomerations of actions by the group members. It is complicated to solve a group decision problem by choosing a single optimal group action. For example, think of a parliament that needs to set up the annual budget. This can be presented as a decision problem of the whole state: “*What should we do with our state money in the upcoming year?*” A *complete* solution to this problem should specify all upcoming annual transactions, which is close to impossible to calculate in advance. *Partial* solutions can give rise to coordination failures, especially when communication is impossible or restricted [12]. Coordination problems have received significant attention in the fields of theoretical economics and philosophy [12, 6, 2, 13]. While researchers often assume that communication is impossible,⁴ in this paper, we will study coordination problems where weak forms of pre-play communication are possible. A strong form of pre-play communication would be where agents can publicly communicate and agree on a group plan [15]. In contrast, we consider situations where agents may be unable to communicate publicly: their messages may reach only some agents, but not all.

We study the necessary and sufficient conditions of such communication to avoid coordination failures. Given an arbitrary coordination problem, who must inform whom about their actions to prevent coordination failures? We present an epistemic game-theoretical framework for addressing these questions. In this framework, we semantically prove our main theorem: the coordination problem can be solved via communication if and only if every subset of agents whose actions are interdependent can engage in semi-private communication within that set.

The paper is organized as follows: in Section 2, we recall some basic definitions and results about coordination games, and fix the terminology and notation. We slightly restate the result of [15]: if we assume that agents can communicate *publicly*, any coordination problem can be solved via the adoption of an optimal, interchangeable plan. In Section 3, we weaken the assumption in terms of who can communicate, and show that the necessary and sufficient communication channels depend on the plan: agents must be able to communicate to avoid coordination failures iff their actions are interdependent. After introducing the formal language and semantics in subsection 3.1, we formally state and semantically prove our main theorem about the necessity and sufficiency of semi-private communication in critical sets in subsection 3.2. In subsection 3.3, we discuss whether semi-private communication is indeed necessary, emphasizing why fully private communication – some agents sharing information, while outsiders do not know that this happens – is not sufficient. We conclude and outline directions for future work in Section 4.

⁴ A notable exception is a paper of [15], where it is assumed that group members are capable of pre-play public communication to establish common knowledge about the adoption of the *plan*.

2 Coordination in deontic games: preliminaries

In this section, we introduce a formal apparatus to be used throughout the paper. We presuppose that the reader is familiar with the concept of *games*.⁵ *Deontic games* are a special class of games in which agents share a common interest and must align their actions to serve that interest [14].⁶ Since multiple joint actions may meet everybody's interest, agents need to coordinate accordingly.

Definition 1 (Deontic games). A deontic game is a tuple $G = (Ags, \{A_i\}, d)$, where Ags is a nonempty finite set of agents; for every $i \in Ags$, A_i is a nonempty finite set of actions available for i ; $d : \prod A_i \rightarrow \{0, 1\}$ is a function that partitions action profiles into deontically ideal ones $d^{-1}(1)$ and deontically non-ideal ones $d^{-1}(0)$. There exists at least one action profile $\alpha \in \prod_{i \in Ags} A_i$, such that $d(\alpha) = 1$. Intuitively, an action profile is deontically ideal iff it promotes the common interest of all agents. We suppose that there is at least one group action that does that.

Given any game $G = (Ags, \{A_i\}_{i \in Ags}, d)$, we use the following notational conventions. $Dom(G) := \prod_{i \in Ags} A_i$, the set of all action profiles will be referred to as a domain of the game. If $Ags = \{1, \dots, n\}$ and $\alpha = \langle a_1, \dots, a_n \rangle \in Dom(G)$, then for any $i \in Ags$: $\alpha(i) := a_i$. Analogously, for any $C \subseteq Ags$ and any $\alpha \in Dom(G)$, $\alpha(C) \in \prod_{i \in C} A_i$ is a projection of α to C , e.g. $\langle a_1, a_2, a_3 \rangle(\{1, 3\}) := \langle a_1, a_3 \rangle$. For any $\alpha, \beta \in Dom(G)$, $C \subseteq Ags$, $\alpha \equiv_C \beta$ iff $\alpha(C) = \beta(C)$.

Example 1. Three people, Anne, Boris and Chloe, live between two train stations: Right and Left. Chloe has severe conflicts with both Anne and Boris. In case she meets either of the two alone, she cannot avoid an argument. Nevertheless, if she meets both Anne and Boris, she will not argue against any of them, because she knows the third party will interfere. Every day Anne, Boris and Chloe have to take a train that stops at both stations. So each of them needs to get to one of the stations to get to that train, without having any preference from which station to depart. All of them want to avoid having arguments at the train stations.

We formalize Example 1 as a deontic game $G = (Ags, \{A_i\}_{i \in Ags}, d)$, where: $Ags = \{a, b, c\}$ is the set of players, Anne, Boris and Chloe, respectively; For every $i \in Ags$, $A_i = \{r_i, l_i\}$: r_i is i 's strategy to go to Left, l_i – to go to Right; d is a deontic ideality function, such that $d(\langle r_a, r_b, l_c \rangle) = d(\langle l_a, l_b, r_c \rangle) = d(\langle r_a, r_b, r_c \rangle) = d(\langle l_a, l_b, l_c \rangle) = 1$, and for any other action profile α , $d(\alpha) = 0$. In other words, it is deontically ideal for Ags that Chloe never meets either Anne or Boris without a third party, hence, she does not start an argument.

	l_a	r_a		l_a	r_a
l_b	1	0		1	0
r_b	0	1		0	1
	l_c			r_c	

Fig. 1. Coordination game for Example 1

Anne may go to any of the two stations only if Boris goes there as well. Analogously, Boris can go to any of the two stations only if Anne comes along. If Boris and Anne cannot communicate to agree upon the station they should go to, there is no way to knowingly avoid the negative outcome.

There are two generally acknowledged ways to prevent coordination failures in the absence of communication: introducing focal points and revising the solution concept. A *focal point* (or *Schelling point*)

⁵ A game is an interactive scenario where the outcomes of individuals' decisions are jointly determined by the choices of all participants. In game theory, players (typically) make rational decisions by selecting from available strategies. Each set of choices, one for each player, generates a distinct outcome with an assigned utility value to each player based on their preferences. For an elaborate introduction to game theory, see any textbook on the topic, e.g., [10].

⁶ In deontic games introduced by [14], agents have a shared moral code rather than a common interest. This difference in interpretations is not essential for our purposes.

is a solution to coordination game that agents tend to choose by default. Thomas Schelling showed the existence of focal points experimentally [11]. Schelling asked his students where and when they would go in case they needed to meet a person in New York City without having a chance to agree on time and place in advance. The most common answer was noon at the Grand Central Terminal. There is no clear reason why this time and place is preferred. This choice is grounded in nothing but the custom of New Yorkers to use Grand Central Terminal as a meeting place. When and how exactly the focal points form is still an open question.

Given that the focal points do not necessarily exist in every case, they do not entirely solve our problem. Sometimes coordination problems can be solved by revised solution concepts. For example, if deontically ideal solutions somehow differ in terms of payoffs, e.g. one Pareto-dominates others⁷, or one of them is less risky⁸, it is reasonable to say that agents are rationally pressured to take this difference into account as well [6]. Another approach is switching to *team reasoning*: first, agents think about what is best for all of them *as a group*, and then individually deliberate on how they should act in order to fulfill their share of the best group behavior [2]. Both approaches are really powerful tools that can solve a wide range of coordination problems, e.g., the stag hunt game and alike. [4] amends team reasoning with a more elaborate participatory-reasoning schema, which can be applied to typical collective action problems. Nonetheless, [4] argues that participatory reasoning may still fail to solve cooperation problems because of external or coordination uncertainty.

If we assume that agents have sources of pre-play communication that allow them to adopt a plan and make it commonly known, then they can agree to narrow their choice between the otherwise equally preferred joint actions [15]. Technically speaking, plan adoption is modeled by updating the ideality function: in the updated game, group actions that go against the plan are no longer deontically ideal.

Definition 2 (Plan updates). *Given a deontic game $G = (Ags, \{A_i\}_{i \in Ags})$, a plan $P \subseteq Dom(G)$ is an arbitrary set of group actions. Then, the update of G with P , denoted as $G \uparrow P$, is the deontic game $G \uparrow P = (Ags, \{A_i\}_{i \in Ags}, d^\uparrow)$, where $d^\uparrow(\alpha) = d(\alpha)$ if $\alpha \in P$; otherwise, $d^\uparrow(\alpha) = 0$.*

As it was shown in [15], agents should adopt *optimal* and *interchangeable* plans, i.e., the set of the group's chosen joint actions should contain only deontically ideal joint actions and presuppose independence between agents' individual actions.⁹

Definition 3 (Acceptability, optimality, interchangeability). *Given a deontic game $G = (Ags, \{A_i\}_{i \in Ags}, d)$, an individual action $a_i \in A_i$ of agent i is called **acceptable** iff it is compatible with the deontically ideal outcome, i.e., there exists an action profile $\alpha \in Dom(G)$, such that $d(\alpha) = 1$ and $\alpha(i) = a_i$. Analogously, given any subgroup $C \subseteq Ags$, a subgroup action $a_C \subseteq \prod_{i \in C} A_i$ is acceptable iff there exists an action profile $\alpha \in Dom(G)$, such that $d(\alpha) = 1$ and $\alpha(C) = a_C$.*

*A plan $P \subseteq Dom(G)$ is **optimal** iff for every $\alpha \in P$, $d(\alpha) = 1$. A plan $P \subseteq Dom(G)$ is **interchangeable** iff for any $\alpha_1, \alpha_2 \in P$, for any disjoint $A, B \subseteq Ags$, there exists an action profile $\alpha_3 \in P$, such that $\alpha_1 \equiv_A \alpha_3$ and $\alpha_2 \equiv_B \alpha_3$. See Figure 2 for an example of an interchangeable plan.*

As it was shown in [15], given a set of agents $Ags = \{1, \dots, n\}$, a deontic game $G = (Ags, \{A_i\}_{i \in Ags}, d)$ and an optimal and interchangeable plan P , individual actions $a_1 \in A_1, \dots, a_n \in A_n$ are acceptable in $G \uparrow P$ iff $\langle a_1, \dots, a_n \rangle \in P$. In other words, the optimal and interchangeable plan narrows the set of acceptable actions for every agent in such a way that every aggregation of members' acceptable actions is in the plan and deontically ideal. That way, every agent can choose any of her acceptable actions without knowing what others are doing, and the deontically ideal outcome will be reached.

⁷ Given two action profiles α, β , α Pareto dominates β iff 1) no one strictly prefers β to α ; 2) at least one agent strictly prefers α to β .

⁸ Informally, given two action profiles α, β , α risk-dominates β iff when each player assigns some probability to the possibility that the other players might not chose their joint strategy, choosing α yields a higher expected payoff than choosing β .

⁹ The notion of interchangeability was firstly introduced in [8]. See [9] for a detailed technical exploration of the logical properties of this notion.

	1_a	2_a	3_a
1_b	1	1	0
2_b	0	0	0
3_b	1	1	0

Fig. 2. Example of a two-player coordination game with an interchangeable plan. $A_a = \{1_a, 2_a, 3_a\}$, $A_b = \{1_b, 2_b, 3_b\}$, $P = \{\langle 1_a, 1_b \rangle, \langle 1_a, 3_b \rangle, \langle 2_a, 1_b \rangle, \langle 2_a, 3_b \rangle\}$. Note that $P = \{1_a, 2_a\} \times \{1_b, 3_b\}$, where $\{1_a, 2_a\}$ are acceptable for a and $\{1_b, 3_b\}$ are acceptable for b .

The set of deontically ideal actions in Example 1 is not interchangeable: $d(\langle r_a, r_b, r_c \rangle) = d(\langle l_a, l_b, l_c \rangle) = 1$ and $\langle l_a, l_b, l_c \rangle \equiv_{\{b,c\}} \langle r_a, l_b, l_c \rangle \equiv_{\{a\}} \langle r_a, r_b, r_c \rangle$, but $d(\langle r_a, l_b, l_c \rangle) = 0$. If Anne, Boris and Chloe have sufficient sources of pre-play communication to establish common knowledge among them, then they can adopt an interchangeable plan to narrow their choice and avoid coordination failures. For example, they can agree that Anne and Boris should go to Left: $P = (\alpha \in \text{Dom}(G) \mid \alpha(\{a, b\}) = \langle l_a, l_b \rangle)$. P is optimal and interchangeable. In that case, both Anne and Boris know they should go to Left, while Chloe knows that she can go to any station, and as long as everyone follows the plan, Chloe will not meet Anne or Bob without a third party.

The solution of [15] underlines the key point: if agents can establish common knowledge among them via pre-play communication, any coordination problem can be solved. For every coordination game G , there exists at least one optimal and interchangeable plan P . Take any action profile $\alpha \in \text{Dom}(G)$, such that $d(\alpha) = 1$ (by Definition 1, such an action profile exists). Trivially, the singleton plan $P = \{\alpha\}$ is optimal and interchangeable. The solution of [15] requires a very strong assumption about agents' ability to communicate. Namely, [15] consider the adoption of the group's plan to be *public*, which results in the plan being commonly known among all members. As they note, "[t]hese idealizing assumptions [about public adoption of the plan] are similar to those that underlie the logic of public announcements" [15, Footnote 14]. Therefore, for a group of agents to adopt the plan, they should be able to communicate *publicly*: agents' messages should simultaneously reach everyone, everyone gets notified that the message has reached anyone, and so on.

3 Weakening the announcements

So far, we have explored solving coordination problems under two assumptions about agents' ability for pre-play communication. If we assume that agents cannot communicate at all, the coordination problem can be solved if there exists a focal point [12] or if deontically ideal outcomes differ in terms of pay-offs or riskiness [6, 2]. On the contrary, if we assume that agents can communicate publicly, they can solve any coordination problem by adopting an optimal interchangeable plan [15]. The first assumption seems too weak: most of the time, group members can share at least some information with at least some of their peers. On the contrary, the second assumption is too strong: rarely can agents publicly announce what strategies they have decided to follow. Often, only some agents have sources of communication, and their messages can reach only some subset of group members. In this section, we relax the assumption of public communication. We show in which cases only private communication inside some subgroups is enough to prevent coordination failures.

As it is shown by [15], there is no need for agents to communicate iff the set of deontically ideal outcomes is interchangeable, or, in other words, if they can achieve deontically ideal states by acting independently from one another. We show that, given any deontic game, it is necessary and sufficient that only agents whose actions are dependent on one another can sequentially let one another know what they are about to do.

To illustrate our point, we go back to Example 1. Intuitively, the actions of Anne and Boris are interdependent: to achieve a deontically ideal outcome, whatever station Anne chooses, Boris should choose the

same station as well, and vice versa. On the contrary, Chloe's actions are independent of the joint actions of Anne and Boris. Chloe's choice of action does not restrict Anne and Boris in their choices, and vice versa. Therefore, it suffices, e.g., for Anne to let Boris know which station she will go to. Suppose Anne privately tells Boris she will go to Left. Then, Boris will know that he should choose Left as well, otherwise, either Anne or he will meet Chloe without a third party. Chloe can safely choose any station she prefers, and since both Anne and Boris will be at Left, Chloe will either meet neither or both of them. Then, the plan of three agents is not interchangeable, but coordination failure can be avoided via private communication. The example indicates that public communication may not always be necessary.

We want to formally verify our intuition about the necessity and sufficiency of private communication between interdependent agents. Therefore, we need to reason about not only (in)dependence between all agents and their common knowledge, as in the case of publicly adopted interchangeable plans, but also about (in)dependence and common knowledge inside subgroups as well. Therefore, first, we define a formal language that is rich enough to express these notions. Then, we define a semantical framework – ex-ante epistemic game models – to interpret that formal language. Finally, we prove the theorem about the necessity and sufficiency of private communication between interdependent agents in that language.

3.1 Formal language, semantics

We fix a finite set of agents $Ags = \{1, \dots, n\}$ and finite sets of the corresponding action names, e.g. for any $i \in Ags$, $A_i = \{1_i, \dots, j_i\}$ for some $j \in \mathbb{N}$. We recursively define the language $\mathcal{L}$ as follows:

$$\pi := a_i \mid \neg\pi \mid (\pi \vee \pi)$$

$$\mathcal{L} \ni \varphi := \delta \mid \pi \mid \neg\varphi \mid (\varphi \vee \varphi) \mid \Box\varphi \mid K_i\varphi \mid C_X\varphi \mid [\pi]_X\varphi$$

where $a_i \in A_i$, $i \in Ags$, $X \subseteq Ags$. δ stands for “a deontically ideal outcome is reached”. a_i means that agent i has done the action a_i . $\Box\varphi$ means that it is necessarily true that φ . $K_i\varphi$ reads as “agent i knows that φ ”. $C_X\varphi$ reads as “it is commonly known in X that φ ” (we will abbreviate C_{Ags} as just C).¹⁰ $[\pi]_X\varphi$ reads as “if it is semi-privately announced to the members of X that π , then φ holds”. By semi-private announcement of π to X we mean the following: all members of X establish common knowledge among them that π , while other agents only get to know that members of X have reached common knowledge *whether* φ .¹¹

Note that $[K_i a_j]_X\varphi$, $[C_X\varphi]_X\psi$, $[[\varphi]_X\psi]_Y\theta$ etc. are not well-formed formulas: we only allow announcements about strategies, not about other announcements or agents' knowledge. Notational conventions: we abbreviate $\wedge, \rightarrow, \leftrightarrow$ via $\neg, \vee$ in a standard way, as well as modalities' duals: $\hat{K}_i := \neg K_i \neg$, $\hat{C}_X := \neg C_X \neg$, $\langle \pi \rangle_X \psi := \neg [\pi]_X \neg \psi$. $\hat{K}_i\varphi$ reads as “ i considers it possible that φ ”, $\hat{C}_X\varphi$ – “it is commonly considered possible among X that φ ”, $\langle \pi \rangle_X\varphi$ – “ π is semi-privately announced to X and φ is true”.

In order to interpret this language, we will use *epistemic game models* [1].

Definition 4 (Epistemic game models). Given a deontic game $G = (Ags, \{A_i\}_{i \in Ags}, d)$, an epistemic game model is a tuple $M = (G, W, \{\sim_i\}_{i \in Ags}, \sigma)$, where:

- G is a deontic game;
- W is a non-empty set of possible worlds;
- $\sim_i \subseteq W \times W$ is an equivalence relation on possible worlds, representing agent's knowledge;
- $\sigma : W \rightarrow \text{Dom}(G)$ is a function that takes a world and returns its corresponding outcome of the game.

For brevity, $\Delta := \{w \in W \mid d(\sigma(w)) = 1\}$, i.e., worlds in Δ correspond to the deontically ideal outcomes.

¹⁰ We adopt a mainstream definition of common knowledge as in [7]: it is commonly known that φ iff everyone knows that φ , everyone knows that everyone knows that φ , everyone knows that everyone knows that everyone knows that φ , and so on, ad infinitum. In what follows, we adopt a formal treatment of common knowledge as in [5].

¹¹ To the best of our knowledge, the concept of semi-private (or semi-public) announcement first appears in [3].

Semi-private communication on epistemic game models is modeled via *semi-private updates*:

Definition 5 (Semi-private updates). For any epistemic game model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, a set of worlds $W' \subseteq W$ and a group $X \subseteq \text{Ags}$, $M_{W'}^X = (G, W, \{\sim_i^{(X, W')}\}_{i \in \text{Ags}}, \sigma)$ is an updated model, where:

- G, W, σ are defined just like in M ;
- $\sim_i^{(X, W')} = \begin{cases} \sim_i \cap (W' \times W') & \text{if } i \in X \\ \sim_i & \text{otherwise} \end{cases}$

$M_{W'}^X$ models the state of affairs after it was semi-privately announced to members of X that the outcome of the game lies in W' .

We will work with a certain subclass of epistemic game models, which we call *ex ante* models. Namely, ex ante models depict the knowledge of agents in games of complete information that they have before any of them has decided upon their strategy. In other words, every agent in ex ante model knows neither her own strategy nor the strategy of any other player, and it is commonly known which strategies every agent has, what outcomes are deontically ideal, and who knows and does not know what.

Definition 6 (Ex ante models). Given any deontic game $G = (\text{Ags}, \{A_i\}_{i \in \text{Ags}}, d)$, an ex ante model for this game is an epistemic game model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, such that:

1. There is a one-to-one correspondence between the domain of the game and the set of possible worlds, i.e. σ is a bijective function;
2. For any agent $i \in \text{Ags}$, $\sim_i = W \times W$.

From now on, we denote the class of ex-ante models as **E**.

Definition 7 (Semantics for $\mathcal{L}$).

For any epistemic model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, let $\sim_X = (\bigcup_{i \in X} \sim_i)^*$ be a reflexive transitive closure of $\bigcup_{i \in X} \sim_i$. Then, for every model M and a world $w \in W$, we inductively define $\models$ relation as follows:

$$\begin{aligned} M, w \models \delta &\Leftrightarrow w \in \Delta & M, w \models \Box \varphi &\Leftrightarrow \forall v \in W : M, v \models \varphi \\ M, w \models a_i &\Leftrightarrow \sigma(w)(i) = a_i & M, w \models K_i \varphi &\Leftrightarrow \forall v \in W (w \sim_i v \Rightarrow M, v \models \varphi) \\ M, w \models \neg \varphi &\Leftrightarrow M, w \not\models \varphi & M, w \models C_X \varphi &\Leftrightarrow \forall v \in W (w \sim_X v \Rightarrow M, v \models \varphi) \\ M, w \models \varphi \vee \psi &\Leftrightarrow M, w \models \varphi \text{ or } M, w \models \psi & M, w \models [\pi]_X \varphi &\Leftrightarrow M, w \models \pi \text{ or } M_{[\pi]}^X, w \models \varphi \end{aligned}$$

where $[\pi] = \{w \in W \mid M, w \models \pi\}$. From now on, we denote the updated models as follows: $M_{\pi}^X := M_{[\pi]}^X = (G, W, \{\sim_i^{X, \pi}\}_{i \in \text{Ags}}, \sigma)$. $M \models \varphi$ iff $M, w \models \varphi$ for all $w \in W$. $\mathbf{E} \models \varphi$ iff $M \models \varphi$ for any ex-ante model M .

After we have defined the semantics for $\mathcal{L}$, we proceed with expressing some crucial notions that we need to formally define necessary and sufficient conditions for communication channels to avoid coordination failures. Given a set of agents $X \subseteq \text{Ags}$, let $A_X = \{\bigwedge_{i \in X} a_i \mid \forall i \in X : a_i \in A_i\}$ be a set of conjunctions of X -members' actions names. Informally, every formula $\varphi \in A_X$ is a name of some joint action of X . We can express the acceptability of (joint) strategies. Namely, given a deontic game G and its ex ante model M a strategy a_i is acceptable iff $M \models \Diamond(a_i \wedge \delta)$, i.e., there exists a deontically ideal world where a_i is done. The acceptability of any joint action $\varphi_X \in A_X$ is expressed analogously: $M \models \Diamond(\varphi_X \wedge \delta)$.

On ex-ante models, we can express *independence* between actions of some subgroups via acceptability. Given any $A, B \subseteq \text{Ags}$, actions of A and B are independent, $A \parallel B$, iff an aggregation of any acceptable joint actions of A and B constitute an acceptable joint action of $A \cup B$:

$$A \parallel B := \bigwedge_{\varphi_B \in A_B} \bigwedge_{\varphi_A \in A_A} (\Diamond(\varphi_A \wedge \delta) \wedge \Diamond(\varphi_B \wedge \delta)) \rightarrow \Diamond(\varphi_A \wedge \varphi_B \wedge \delta) \quad (1)$$

Consequently, we can define interchangeability of the set of deontically ideal action profiles as independence of all disjoint subgroups:

$$\|_{Ags} := \bigwedge_{A \cap B = \emptyset} A \| B$$

Alternatively, given $Ags = \{1, \dots, n\}$,

$$\|_{Ags} := \bigwedge_{a_1 \in A_1} \dots \bigwedge_{a_n \in A_n} (\Diamond(a_1 \wedge \delta) \wedge \dots \wedge \Diamond(a_n \wedge \delta)) \rightarrow \Box((a_1 \wedge \dots \wedge a_n) \rightarrow \delta) \quad (2)$$

It is a routine to check that both definitions of $\|_{Ags}$ are equivalent. Note that on ex-ante models (or any models, where σ is bijective), there is an equivalent definition of $\|_{Ags}$:

$$\mathbf{E} \models \|_{Ags} \leftrightarrow \left(\bigwedge_{a_1 \in A_1} \dots \bigwedge_{a_n \in A_n} (\Diamond(a_1 \wedge \delta) \wedge \dots \wedge \Diamond(a_n \wedge \delta)) \rightarrow \Diamond(a_1 \wedge \dots \wedge a_n \wedge \delta) \right)$$

since there is a single world, where $a_1 \wedge \dots \wedge a_n$ is true.

We can express agents' knowledge about the acceptability of their actions. Namely, given a pointed model (M, w) , $M, w \models \hat{K}_i(a_i \wedge \delta)$ iff i considers a_i acceptable. Consequently, $M, w \models K_i(a_i \rightarrow \neg \delta)$ means that i knows that her action a_i is unacceptable. Agents' actions may not be independent in general, but if we look only at actions that agents themselves consider acceptable, they might well be independent.

Given a pointed epistemic game model (M, w) of a deontic game G , we say that G is *epistemically interchangeable* at (M, w) iff every agent considers it possible to fulfill the plan and the agglomeration of any actions that agents themselves consider acceptable is acceptable. Formally:

$$\|_{Ags}^{\sim} := \bigwedge_{i \in Ags} \hat{K}_i \delta \wedge \bigwedge_{a_1 \in A_1} \dots \bigwedge_{a_n \in A_n} (\hat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \hat{K}_n(a_n \wedge \delta)) \rightarrow \Box((a_1 \wedge \dots \wedge a_n) \rightarrow \delta) \quad (3)$$

In the case of epistemic interchangeability, a deontically ideal outcome will be reached without additional communication, given that no one will do an action that she knows to be unacceptable. If the agents have achieved epistemic interchangeability, even if the set of deontically ideal outcomes is not interchangeable, they can use their knowledge to act *as if* that set was interchangeable. They can just avoid doing actions that they know to be unacceptable, and some deontically ideal outcome will inevitably be reached. Note that on ex-ante models, just like on any other models with bijective σ , $\|_{Ags}^{\sim}$ has an equivalent definition:

$$\mathbf{E} \models \|_{Ags}^{\sim} \leftrightarrow \left(\bigwedge_{i \in Ags} \hat{K}_i \delta \wedge \bigwedge_{a_1 \in A_1} \dots \bigwedge_{a_n \in A_n} (\hat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \hat{K}_n(a_n \wedge \delta)) \rightarrow \Diamond(a_1 \wedge \dots \wedge a_n \wedge \delta) \right)$$

since there is a unique world, where $a_1 \wedge \dots \wedge a_n$ is true.

It is straightforward that from $M, w \not\models A \| B$ it follows that $M, w \not\models (A \cup X) \| (B \cup Y)$ for any $X, Y \subseteq Ags$. In other words, dependence is *monotonic*: as soon as actions of two subgroups depend on one another, actions of any of their supergroups are dependent as well. If we want to capture the genuine, irreducible dependence between subgroups, we need the notion of a *critical set*.¹² Informally, $X \subseteq Ags$ is a critical set iff there is a dependence that involves all members of X .

Definition 8 (Critical sets in deontic games). *Given an ex-ante model of some deontic game M , a set $X \subseteq Ags$ is called critical iff there is a critical partition of X onto X_1 and X_2 , i.e. a partition such that:*

1. $M \not\models X_1 \| X_2$, i.e., X_1 and X_2 are not independent;
2. $M \models (X_1 \cap B) \| (X_2 \cap B)$ for any $B \subset X$, i.e., dependence between X_1 and X_2 cannot be reduced to the dependence between their proper subgroups.

¹² [9] introduce the notions of critical sets and critical partitions for purely technical reasons, to establish a completeness proof for their logic.

Actions of a critical set's members are interdependent: a good choice of strategy of any member is restricted by the choices of all other members.

As for semi-private announcements, we say that an announcement $[\pi]_X$ is *internal* iff π is a formula that is built only from action names $\{A_i\}_{i \in X}$. In other words, $[\pi]_X$ is an internal semi-private announcement iff only information about strategies of some members of X is announced in X . An announcement $[\pi]_X$ is *individual* iff π is built only from action names A_i of some single agent $i \in \text{Ags}$ and Boolean connectives, i.e., $[\pi]_X$ is *individual* iff only information about actions of one agent is announced. Assume $i \in X \subseteq \text{Ags}$ and π is a formula built from A_i and Boolean connectives. Then, $[\pi]_X$ is an *internal individual semi-private announcement*. Such announcements model semi-private messages of i to X about i 's choice of strategy: it might be the exact choice of i 's strategy or some partial solution.

By semi-private *disclosure of i 's strategy to X* we mean the announcement, after which i 's strategy is commonly known in X , while outsiders only get to know that agents in X get to know the exact strategy of i , whatever this strategy is. Such disclosures are also expressible in our language:

Definition 9 (Semi-private disclosure of one's strategy). *Given a deontic game $G = (\text{Ags}, \{A_i\}_{i \in \text{Ags}}, d)$, its ex ante model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$ and a world $w \in W$, the semi-private disclosure of i 's strategy to X is the semi-private announcement of i 's strategy to X , i.e. $[\sigma(w)(i)]_X$, followed by semi-private announcements of negations of all other i 's strategies to X , i.e. $[\neg i_j]_X$ for any $i_j \in A_i \setminus \{\sigma(w)(i)\}$. For example, if $A_i = \{i_1, i_2, i_3\}$ and $\sigma(w)(i) = i_1$, i.e. i does i_1 in w , then the semi-private disclosure of i 's strategy to X in w is the following announcement: $\langle i_1 \rangle_X \langle \neg i_2 \rangle_X \langle \neg i_3 \rangle_X$. It is a routine to check that this announcement indeed works as a semi-private disclosure of i 's strategy to X , i.e., that:*

$$M, w \models \langle i_1 \rangle_X \langle \neg i_2 \rangle_X \langle \neg i_3 \rangle_X C \left(\bigwedge_{i_j \in A_i} i_j \rightarrow C_X i_j \right)$$

From now on, we will use the abbreviation $M, w \models [i]_X \phi$ for a semi-private disclosure of i 's strategy $\sigma(w)(i)$ to X .

Returning to Example 1, we can now formally demonstrate how pre-play semi-private communication can solve coordination problems. Let $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$ be the ex-ante model for the deontic game for Example 1, presented in Section 2. Note that the set $\{a, b\}$ is critical: $M \not\models a \parallel b$, while trivially $M \models a \parallel \emptyset$ and $M \models \emptyset \parallel b$.

Without loss of generality, we can assume that Anne goes to Right, while Boris and Chloe go to Left. Could they have avoided this suboptimal outcome by pre-play communication?

Let $\sigma((r, l, l)) = w$ be a real world. Suppose Anne had semi-privately disclosed her action to Boris. Then, Anne would still consider going Right acceptable and no longer considers it possible to go left:

$$M, w \models [a]_{\{a, b\}} K_a \neg l_a \wedge \widehat{K}_a(r_a \wedge \delta)$$

Then, Boris will consider the only strategy of his acceptable, to go to the same station as Anne, i.e., Right. Knowing where Anne will go, Boris knows that if he does not go to the same station, either he or Anne will meet Chloe without a third party, which is deontically non-ideal:

$$M, w \models [a]_{\{a, b\}} (K_b(l_b \rightarrow \neg \delta) \wedge \widehat{K}_b(r_b \wedge \delta))$$

As a result, after the semi-private disclosure, both Anne and Boris will consider only a single strategy of each of them acceptable – to go to Right – while Chloe will still consider both of her strategies acceptable. Any combination of Chloe's strategy and the joint strategy of Anne and Boris that they consider acceptable is acceptable, i.e., after the announcement, agents will reach epistemic interchangeability: $M, w \models [a]_{\{a, b\}} \widetilde{\parallel}_{\text{Ags}}$.

Chloe will not know what strategy Anne has disclosed to Boris, but since the disclosure was semi-private, she will learn that Anne and Boris know enough to choose the same train station. Therefore, she

will get to know that all of them have reached epistemic interchangeability. Moreover, after the semi-private disclosure of Anne's strategy, it will be commonly known: $M, w \models [a]_{\{a,b\}} C(\|\widetilde{\sim}_{Ags}\rangle)$. Therefore, it is commonly known that everyone knows enough to act as if the set of deontically ideal outcomes was interchangeable, i.e., it is commonly known that agents can avoid coordination failure by choosing strategies they consider acceptable, and no further communication is needed.

3.2 When semi-private communication is enough: formal proof

In this section, we prove our main theorem: semi-private communication via internal individual announcements in critical sets is a necessary and sufficient condition for solving any coordination problem. We limit communication to *internal individual* semi-private announcements to keep things individualistic. Announcements should be individual to exclude a vicious circle in our explanation: if we allow groups to announce their choice of group actions, we will need to explain how these groups solve coordination problems to make such truthful announcements. Moreover, the announcements should be internal, so that we exclude cases of mind-reading: only the agent herself can announce what she has chosen to do. First, we show the necessity of such communication. In other words, we need to show that we cannot reach the common knowledge of epistemic interchangeability by a sequence of announcements if such announcements do not reach at least one member of at least one critical set.

Theorem (Communication in critical sets is necessary). Take any game $G = (Ags, \{A_i\}_{i \in Ags}, d)$ with its ex ante game model $M = (G, W, \{\sim_i\}_{i \in Ags}, \sigma)$ and any possible world $w \in W$. Take any sequence of individual internal semi-private announcements $[\alpha_1]_{X_1} \dots [\alpha_n]_{X_n}$, such that $M, w \models \alpha_1 \wedge \dots \wedge \alpha_n$ and there is a critical set $X \subseteq Ags$, s.t. $X \not\subseteq \bigcup_{1 \leq i \leq n} X_i$. Then, $M, w \not\models \langle \alpha_1 \rangle_{X_1} \dots \langle \alpha_n \rangle_{X_n} C(\|\widetilde{\sim}_{Ags}\rangle)$.

Proof. See Theorem 1 in Appendix.

Now we will show that communication inside only critical sets is sufficient. To make our formulation precise and justify it, we point out that epistemic interchangeability presupposes that everyone considers it possible to reach a deontically epistemic state. Therefore, to reach the common knowledge of epistemic interchangeability, agents should announce they are about to do actions that they consider acceptable, given what they know. Going back to Example 1: if Anne announces that she is to go Left, Boris cannot announce that he goes to Right to solve the coordination problem: if such announcements are truthful, then Anne and Boris will know they will not reach an epistemically ideal state. Having this in mind, we formulate the sufficiency condition as follows:

Theorem (Communication in critical sets is sufficient). Take any deontic game $G = (Ags, \{A_i\}_{i \in Ags}, d)$ with its ex ante epistemic game model $M = (G, W, \{\sim_i\}_{i \in Ags}, \sigma)$. Let $\mathcal{X} = \{X_1, \dots, X_n\}$ be the collection of all critical sets of G . Then, there exists a sequence of individual internal semi-private announcements $[\alpha_1]_{X_1} \dots [\alpha_q]_{X_n}$ such that:

1. For every critical set $X_i \in \mathcal{X}$, there is at least one corresponding announcement $[\alpha_i]_{X_i}$, and for any set Y that is not critical there is no announcement $[\alpha]_Y$;
2. At every m th step of the sequence $[\alpha_1]_{X_1} \dots [\alpha_m]_{X_j}$, if α_{m+1} is an expression about actions of $i \in Ags$, $M \models [\alpha_1]_{X_1} \dots [\alpha_m]_{X_j} \widetilde{K}_i(\alpha_{m+1} \wedge p)$, i.e. at every step the announced action of the agent is considered acceptable by that agent after all the announcements that were already made.
3. $M \models [\alpha_1]_{X_1} \dots [\alpha_n]_{X_n} C(\|\widetilde{\sim}_{Ags}\rangle)$, i.e., after the sequence of the announcements, the common knowledge about epistemic interchangeability is established.

Proof. See Theorem 2 in Appendix.

Interestingly, the result of [15] follows from Theorems 1,2. If the commonly known plan is already interchangeable, then the only critical set is $\emptyset$, therefore, the absence of communication between agents is sufficient to solve the coordination problem and there is no necessity for any agents to communicate.

3.3 Discussion: would full privacy be sufficient?

We wonder: what would be different in the scenario of Example 1 if Anne had communicated privately to Boris, unbeknownst to Chloe? She still could have done her share – just come to any station – without knowing that others have enough information to independently do their shares. So, one might argue that private disclosures suffice (in contrast with semi-private disclosures). Is it necessary to achieve common knowledge that everyone has enough information? Or is it enough to achieve the state where everyone actually has enough information, even if unknown to others? We can find good arguments in favour of the necessity of common knowledge in the literature on *group identification*. In game-theoretic scenarios, agents can reason either individualistically, i.e. ask themselves “*what should I do?*” or engage in *team reasoning*, i.e. ask themselves “*what we as a group should do?*” In the second scenario, the individual choices of group members’ strategies should follow from the choice of the optimal joint strategy of the group. Nevertheless, it is not always obvious which mode of practical reasoning one should employ. As pointed out by [13], often one should engage in team reasoning only if one is sure that others identify themselves as group members and reason “socially” as well. In case one lacks reasons to believe that other group members are engaged in team reasoning, one may lose the commitment to act towards the common good.

We can make a similar point with respect to (absence of) sufficient communication. Some deontic games require agents to communicate, ergo one may lose their commitment to promote common interest in case they lack reasons to believe that others communicate in the way that the game requires them to do. Although Example 1 is insufficiently rich to illustrate this, see Example 2 in the Appendix for an elaborate illustration of such a case.

4 Conclusion

In this paper, we have explored the solutions to coordination problems via pre-play communication. Namely, we have specified the necessary and sufficient conditions of communication, via which group members can prevent coordination failures: it is semi-private communication inside the critical sets. Our findings indicate that the notion of acceptable action may not be sharp enough: the action is acceptable iff it leaves other agents with causal, but not epistemic ability to reach the deontically ideal outcome. We have left some questions untouched, inviting further research.

We have figured out *who* should be able to communicate to solve the coordination problem, but we have not specified *what* should be communicated. So far, we have allowed any individual internal semi-private announcements: agents are free to disclose any truthful information about their own actions. This freedom follows from the idealization that every agent is omniscient with respect to her own actions: every agent can freely decide what she is to do and disclose it to anyone she can communicate with. This idealization should be relaxed. We adopt partial plans exactly because sometimes it is impossible or too costly for us to settle upon the exact action we are to take. Some plans may pressure agents to disclose their exact strategy to avoid coordination failures, while these agents may lack the resources to decide upon their exact strategy. Such restrictions should also be taken into account.

Finally, we have ignored the order of announcements, while the first agent to announce her strategy restricts the acceptable choices of others, which may result in some advantage to her. This obstacle points out that we need to explore *priority orders* among group members that often takes place in such scenarios.

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A Example for Subsection 3.3

Example 2. Chloe supervises the research of two students: Anne and Boris. All three are involved in a project that Chloe can either fully write by herself and submit before the deadline (p_c) or let her students practice the research and write one proof, hence, wait till they finish and submit the paper as co-authored after (n_c). To write the proof, both Anne and Boris should either simultaneously do some model theory (p_a, p_b) or proof theory (n_a, n_b): in both cases, they will finish the proof. Obviously, if Anne and Boris work on the proof while Chloe finishes the whole paper by herself and submits it, Anne and Boris will get no credit for the work, unlike Chloe. In parallel, Anne and Boris have managed to discover a minor improvement to the part written entirely by Chloe. If they show it to Chloe, she will have to mention both of them in the erratum of the paper, which will give them less credit than being co-authored. To explicitly show that improvement, Anne should do some model theory, and Boris should do some proof theory (p_a, n_b). But this is worth working on, only if Chloe is to write the whole paper by herself: otherwise, if Chloe chooses to wait for Anne and Bob to write the proof, while they work on the minor improvement, the paper will not be finished until the deadline, and will not be published.

Consider the following three-players game $G = (Ags, \{A_i, u_i\}_{i \in Ags})$, where $Ags = \{a, b, c\}$ represents Anne, Boris and Chloe, respectively; $A_i = \{p_i, n_i\}$ for any $i \in Ags$. For Anne and Bob, p_a and p_b mean Anne resp. Bob works on model theory, n_a and n_c – Anne resp. Bob works on proof theory. For Chloe, p_c – to write the whole paper by herself, n_c – wait for Anne and Bob; payoffs are specified as in Figure 3.

	p_a	n_a		p_a	n_a
p_b	(0, 0, 1)	(0, 0, 1)	p_b	(1, 1, 1)	(0, 0, 0)
n_b	(0.9, 0.9, 1)	(0, 0, 1)	n_b	(0, 0, 0)	(1, 1, 1)
	p_c			n_c	

Fig. 3. Three-player game G .

The set of deontically ideal joint actions of Ags is $P = \{(p_a, p_b, n_c), (n_a, n_b, n_c)\}$. These action profiles yield the highest payoffs for all agents: the strongest result is published, and all three get credits for it. Let us reason about the game from the position of c . Ideally, c should commit to n_c , i.e., to not writing the whole paper by her own. If a and b will not fulfill their joint share, i.e., will do either (p_a, n_b) or (n_a, p_b) , c will get 0: if Anne and Boris work on something else, while she waits for their contribution, Chloe will not publish anything. On the other hand, c can secure her own pay-off if she simply ignores Anne and Boris and chooses p_c : in such a case, she will get 1 no matter what a and b are going to do. Therefore, it makes sense for c to adopt team reasoning and commit to n_c only if she is sure that a and b coordinate their actions to do either (p_a, p_b) or (n_a, n_b) . As we have shown in the previous sections, this requires communication between a and b . Therefore, if c does not know if a and b communicate, c may lack reasons to believe that a and b will fulfill their share of the plan, and lose her commitment to do n_c .

Let us look at the situation from the point of a and b . Assume they do communicate and hence commonly know that they have epistemic ability to bring about that (p_a, p_b) or (n_a, n_b) , i.e., to write the proof Chloe has delegated to them. Yet they know that if c does not know about their communication, then c is likely to lose her commitment to n_c and will probably avoid the risk by writing the paper by herself. But if Chloe does so, then it is in the interest of a and b to choose p_a and n_b respectively (i.e., to work on the minor improvement) to secure the payoffs 0.9 for each. The joint action (p_a, n_b) is unacceptable. Therefore, if a and b do not know that c knows that they have shared enough information to coordinate their actions, they may lose their commitment to promote the common interest. We can go on and draw the same conclusion from the failure of higher-order knowledge of c and a with b about the communication between a and b .

B Proofs

Lemma 1 (Dependencies are reducible to critical sets). *For any ex-ante model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$ and two disjoint subsets $A, B \subseteq \text{Ags}$, if $M \not\models A \parallel B$, then there is a critical partition C_1, C_2 , such that $C_1 \subseteq A$ and $C_2 \subseteq B$.*

Proof. See proof of Lemma 3 in [9].

Lemma 2. *For any ex-ante model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, $M \not\models A \parallel B$ for any non-trivial (but not necessarily critical) partition A, B of a critical set C .*

Proof. See Lemma 2 in [9, p.67].

Lemma 3. *Given a coordination game $G = (\text{Ags}, \{A_i\}_{i \in \text{Ags}}, d)$, it ex-ante model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$ and any agent $i \in \text{Ags}$, either $M \models i \parallel \bar{i}$ or there exists a critical set C , such that $i \in C$ and $\{i\}, C \setminus \{i\}$ is a critical partition.*

Proof. Take any agent $i \in \text{Ags}$ and assume that $M \not\models i \parallel \bar{i}$. Since Ags is finite, so $\bar{i}$ is. Note that trivially $M \models i \parallel \emptyset$. Hence, there exists a (not necessarily unique) $\subseteq$ -minimal non-empty set $B \subseteq \bar{i}$, such that $M \not\models i \parallel B$. Fix such $B \subseteq \bar{i}$. We claim that $B \cup \{i\}$ is a critical set and $B, \{i\}$ is a critical partition. By our assumption, $M \not\models i \parallel B$. Moreover, since B is a $\subseteq$ -minimal subset of $\bar{i}$ which is not independent from i , for any proper subset $C \subset B$, $M \models i \parallel C$. Take any subset $D \subset B \cup \{i\}$. If $i \in D$, then $\{i\} \cap D = \{i\}$ and $D \cap B = C$ for some $C \subset B$. In such a case, $M \models \{i\} \cap D \parallel B \cap D$, since it is equivalent to $M \models i \parallel C$, where C is some proper subset of B , and B is a $\subseteq$ -minimal set, which is not independent from i . If $i \notin D$, then $M \models \{i\} \cap D \parallel B \cap D$, since it is equivalent to $M \models \emptyset \parallel B \cap D$, which holds trivially.

Lemma 4 (Epistemic independence is preserved under individual announcements).

For any $X \subseteq \text{Ags}$, let $\|\tilde{X}\| := (\bigwedge_{i \in X} \bigwedge_{a_i \in A_i} \widehat{K}_1(a_i \wedge \delta)) \rightarrow \Diamond(\bigwedge_{i \in X} \bigwedge_{a_i \in A_i} a_i \wedge \delta)$. Given any ex ante epistemic game model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, any world $w \in W$ and any sequence of individual announcements $[\pi_1]_{X_1}, \dots, [\pi_n]_{X_n}$, if $M, w \models \|\tilde{X}\|$ for some $X \subseteq \text{Ags}$, then $M, w \models [\pi_1]_{X_1} \dots [\pi_n]_{X_n}(\|\tilde{X}\|)$.

Proof. W.l.o.g., let $X = \{1, \dots, n\}$ for some $n \in \mathbb{N}$. Assume that $M, w \models \|\tilde{X}\|$, i.e. for any $a_1 \in A_1, \dots, a_n \in A_n$, $M, w \models (\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_n(a_n \wedge \delta)) \rightarrow \Diamond(a_1 \wedge \dots \wedge a_n \wedge \delta)$.

For brevity, let $M_1 = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$ be a model that we get after the announcements $[\pi_1]_{X_1}, \dots, [\pi_n]_{X_n}$ in M . Let $!A_i = \{a_i \in A_i \mid \exists v \in W : w \sim_i v \text{ and } M, v \models a_i \wedge \delta\}$ be a set of actions that i still considers acceptable after the announcements. Note that for every agent $i \in \text{Ags}$, $!A_i \subseteq A_i$. Take any $b_1 \in !A_1, \dots, b_n \in !A_n$. By definition, $M_1, w \models \widehat{K}_1(b_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_n(b_n \wedge \delta)$. Since $b_1 \in !A_1, \dots, b_n \in !A_n$ and $!A_1 \subseteq A_1, \dots, !A_n \subseteq A_n$, $b_1 \in A_1, \dots, b_n \in A_n$. Then, by our initial assumption, $M, w \models \Diamond(b_1 \wedge \dots \wedge b_n \wedge \delta)$. Then, there exists a world $v \in W$, such that $v \in \Delta$ and $\sigma(v)(i) = b_i$ for every $i \in \{1, \dots, n\}$. Since the set of possible worlds W and the function σ in the updated model M_1 are the same, $M_1 \models \Diamond(b_1 \wedge \dots \wedge b_n \wedge \delta)$. Hence, $M_1, w \models \Diamond(b_1 \wedge \dots \wedge b_n \wedge \delta)$. Given our assumption that b_i is an arbitrary member of $!A_i$ for every $i \in X$, $M_1, w \models (\widehat{K}_1(b_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_n(b_n \wedge \delta)) \rightarrow \Diamond(b_1 \wedge \dots \wedge b_n \wedge \delta)$ for every $b_1 \in A_1, \dots, b_n \in A_n$ (if $b_i \notin !A_i$ for some $i \in X$, then the implication is vacuously true). In other words, $M_1, w \models \|\tilde{X}\|$. Then, by Def. 7, $M, w \models [\pi_1]_{X_1} \dots [\pi_n]_{X_n}(\|\tilde{X}\|)$.

Corollary 1. *If $M, w \not\models [\alpha_1]_{X_1}, \dots, [\alpha_n]_{X_n}(\|\tilde{X}\|)$, then $M, w \not\models \|\tilde{X}\|$.*

Lemma 5. *Given any ex ante model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, if $M, w \models \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$ for some $i \in \text{Ags}$, $\{1, \dots, j\} \subseteq \text{Ags}$, s.t. $i \notin \{1, \dots, j\}$, then for any internal semi-private announcement $[\pi]_{\{1, \dots, j\}}$:*

$$M, w \models [\pi]_{\{1, \dots, j\}} \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$$

Proof. Assume that $M, w \models \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$. If $M, w \not\models \pi$, then vacuously $M, w \models [\pi]_{\{1, \dots, j\}} \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$.

Let $M, w \models \pi$. Then, since $M, w \models \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$, there exists $w_1, \dots, w_n$ and $v_1, \dots, v_j \in W$, such that $w \sim_i w_1, \dots, w \sim_i w_j$, $w_1 \sim_1 v_1, \dots, w_n \sim_j v_j$ and $M, v_1 \models \delta \wedge a_1, \dots, M, v_j \models \delta \wedge a_j$. Obviously, $\sim_l^{(X, \pi)}$ is reflexive for any $l \in \text{Ags}$. Then, $w \sim_l^{(X, \pi)} w$, $v_1 \sim_l^{(X, \pi)} v_1, \dots, v_j \sim_l^{(X, \pi)} v_j$ for any $l \in \text{Ags}$. Then, $M_\pi^X, v_1 \models \widehat{K}_1(a_1 \wedge \delta), \dots, M_\pi^X, v_j \models \widehat{K}_j(a_j \wedge \delta)$.

Since $i \notin \{1, \dots, j\}$, $\sim_i^{(X, \pi)} = \sim_i = W \times W$. Therefore, $w \sim_i^{(X, \pi)} v_1, \dots, w \sim_i^{(X, \pi)} v_n$. Consequently, $M_\pi^X, w \models \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$, i.e. $M, w \models [\pi]_X \widehat{K}_i(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_j(a_j \wedge \delta))$.

Theorem 1 (Communication in critical sets is necessary).

For any ex ante epistemic game model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$, any possible world $w \in W$ and any sequence of individual internal semi-private announcements $[\pi_1]_{X_1} \dots [\pi_n]_{X_n}$, such that $M, w \models \pi_1 \wedge \dots \wedge \pi_n$ and there is a critical set $X \subseteq \text{Ags}$, s.t. $X \not\subseteq \bigcup_{1 \leq i \leq n} X_i$: $M, w \not\models \langle \pi_1 \rangle_{X_1} \dots \langle \pi_n \rangle_{X_n} C(\|\widetilde{\sim}_{\text{Ags}})$.

Proof. Take any deontic game $G = (\text{Ags}, \{A_i\}_{i \in \text{Ags}}, d)$ and its ex ante game model $M = (G, W, \{\sim_i\}_{i \in \text{Ags}}, \sigma)$. Take any world $w \in W$ and any critical set $X \subseteq \text{Ags}$. Assume any sequence of individual internal semi-private announcements $[\pi_1]_{X_1} \dots [\pi_n]_{X_n}$, such that $M, w \models \pi_1 \wedge \dots \wedge \pi_n$ and $X \not\subseteq \bigcup_{1 \leq i \leq n} X_i$. In other words, $[\pi_1]_{X_1} \dots [\pi_n]_{X_n}$ is a sequence of truthful announcements in subgroups that does not cover X , i.e. at least some agents in X are outsiders for all the announcements in the sequence. For brevity, let $X = \{1, \dots, l\}$, $X \cap \bigcup_{1 \leq i \leq n} X_i = Y = \{1, \dots, k\}$ for some $k < l$. Agents $\{i \in \text{Ags} \mid k+1 \leq i \leq l\}$ are members of the critical set X that are outsiders for all the announcements that were made. Since outsiders do not learn any new factual information, for every $j \in X \setminus Y$, $M, w \models \widehat{K}_j(a_j \wedge \delta) \leftrightarrow \langle \pi_1 \rangle_{X_1} \dots \langle \pi_n \rangle_{X_n} \widehat{K}_j(a_j \wedge \delta)$. Since X is a critical set, from Lemma 2 it follows that $M \not\models Y \parallel X \setminus Y$, i.e. there exist some joint strategies $\bigwedge_{i \in Y} a_i \in A_Y, \bigwedge_{j \in X \setminus Y} a_j \in A_{X \setminus Y}$, such that $M, w \models \Diamond(\bigwedge_{i \in Y} a_i \wedge \delta) \wedge \Diamond(\bigwedge_{j \in X \setminus Y} a_j \wedge \delta) \wedge \Box((\bigwedge_{i \in Y} a_i \wedge \bigwedge_{j \in X \setminus Y} a_j) \rightarrow \neg \delta)$, i.e. there is some acceptable joint action of Y and some acceptable joint action of $X \setminus Y$, such that the aggregation of these two joint actions is unacceptable. Agents outside Y do not know what exactly agents inside Y have learned, and consider it possible that agents inside Y still consider $\bigwedge_{1 \leq i \leq k} a_i$ acceptable. In other words, by Lemma 5 it follows that $M, w \models \langle \alpha_1 \rangle_{X_1} \dots \langle \alpha_n \rangle_{X_n} \widehat{K}_j(\widehat{K}_1(a_1 \wedge p) \wedge \dots \wedge \widehat{K}_k(a_k \wedge p))$ for every $j \in X \setminus Y$. As they, as outsiders, still consider the same actions of themselves acceptable, every outsider considers $\bigwedge_{k+1 \leq j \leq l} a_j$ acceptable. Therefore, for every $j \in X \setminus Y$, we get the next picture:

$$M, w \models \langle \pi_1 \rangle_{X_1} \dots \langle \pi_n \rangle_{X_n} \widehat{K}_j(\widehat{K}_1(a_1 \wedge \delta) \wedge \dots \wedge \widehat{K}_l(a_l \wedge \delta) \wedge \neg \Diamond(a_1 \wedge \dots \wedge a_l \wedge \delta))$$

which means that $M, w \models \langle \pi_1 \rangle_{X_1} \dots \langle \pi_n \rangle_{X_n} \widehat{K}_j(\neg \|\widetilde{\sim}_{\text{Ags}})$, therefore, $M, w \not\models C(\|\widetilde{\sim}_{\text{Ags}})$. We can conclude that after any sequence of individual internal semi-private announcements that do not cover all critical sets, we will not reach the state where it is commonly known that the set of deontically ideal outcomes is epistemically interchangeable. Hence, communication that covers critical sets is a necessary condition of reaching such a state.

Theorem 2 (Communication in critical sets is sufficient).

Take any deontic game G with its ex ante epistemic game model M . Let $\mathcal{X} = (X_1, \dots, X_n)$ be a collection of all critical sets of G . Then, there exists a sequence of individual internal semi-private announcements $[\pi_1]_{X_1} \dots [\pi_q]_{X_n}$ such that:

1. For every critical set $X_i \in \mathcal{X}$, there is at least one corresponding announcement $[\pi_i]_{X_i}$, and for any set Y that is not critical there is no announcement $[\pi]_Y$;
2. At every m th step of the sequence $[\pi_1]_{X_1} \dots [\pi_m]_{X_j}$, if π_{m+1} is an expression about actions of $i \in \text{Ags}$, $M, w \models [\pi_1]_{X_1} \dots [\pi_m]_{X_j} \widehat{K}_i(\pi_{m+1} \wedge \delta)$, i.e. at every step the announced action of the agent is considered acceptable by that agent after all the announcements that were already made.
3. For every world $w \in W$, $M, w \models [\pi_1]_{X_1} \dots [\pi_q]_{X_n} C(\|\widetilde{\sim}_{\text{Ags}})$.

Proof. W.l.o.g., assume $\mathcal{X} = \{X_1, \dots, X_n\}$ and $X_i = \{x_i^1, \dots, x_{j_i}^1\} \subseteq \text{Ags}$ for every $i \in \{1, \dots, n\}$, and $A_i = \{0, 1\}$ for every $i \in \text{Ags}$.¹³ Take any deontically ideal world, i.e., $w \in W$, such that $M, w \models \delta$. For brevity, for any $j \in \text{Ags}$ let $w_j := \sigma(w)(j)$ be the j 's action in w . Then, we construct a sequence, where for every critical set $X_i \in \mathcal{X}$, every member of X_i semi-privately discloses her strategy at w to all members of X_i :

$$[\mathcal{X}] := [w_{x_1^1}]_{X_1} \dots [w_{x_{j_n}^1}]_{X_n}$$

For brevity, let $M_! = (G, W, \{\sim_i^!\}_{i \in \text{Ags}}, \sigma)$ be a model that we get after performing the sequence of announcements $[\mathcal{X}]$ in M .

We claim that $[\mathcal{X}]$ satisfies all three requirements. (1) is trivially satisfied, given that $\mathcal{X} = \{X_1, \dots, X_n\}$. (2) is satisfied as well. Take any world $v \in W$ and the m th step of the sequence $[\mathcal{X}]$ for some arbitrary $m \in \mathbb{N}$ that is less or equal to the length of $[\mathcal{X}]$. Assume that m th step of the sequence is $[w_{x_l^m}]_{X_m}$ and $j+1$ th step is $[w_{x_{l+1}^m}]_{X_m}$. If $M, v \not\models w_{x_1^m} \wedge \dots \wedge w_{x_l^m}$, then vacuously $M, v \models [w_{x_1^m}]_{X_1} \dots [w_{x_l^m}]_{X_m} \hat{K}_{x_{l+1}^m} (w_{x_{l+1}^m} \wedge p)$. Assume that $M, v \models w_{x_1^m} \wedge \dots \wedge w_{x_l^m}$. Then, note that $M, w \models w_{x_1^m} \wedge \dots \wedge w_{x_l^m}$, i.e. in both w and v all the propositions that are announced so far are true. Hence, in the updated model

$$(w, v) \in \sim_j^!$$

for every $j \in \text{Ags}$, while $M_!, w \models (w_{x_{l+1}^m} \wedge \delta)$. Hence, $M, v \models [w_{x_1^m}]_{X_1} \dots [w_{x_l^m}]_{X_m} \hat{K}_{x_{l+1}^m} (w_{x_{l+1}^m} \wedge \delta)$.

It is left for us to prove that $M \models [\mathcal{X}]C(\|\sim_{\text{Ags}})$. We do it by proving that $M, v \models [\mathcal{X}](\|\sim_{\text{Ags}})$ for any $v \in W$. Take any world $v \in W$. If $M, v \not\models w_{x_1^1} \wedge \dots \wedge w_{x_{j_n}^1}$, then $M, v \models [\mathcal{X}](\|\sim_{\text{Ags}})$ holds vacuously. For contradiction, assume that $M, v \models w_{x_1^1} \wedge \dots \wedge w_{x_{j_n}^1}$ and $M_!, v \not\models \|\sim_{\text{Ags}}\|$. Then, by Corollary 1, $M, v \not\models \|\sim_{\text{Ags}}\|$. Since M is an ex ante model, $\hat{K}_i \varphi \leftrightarrow \Diamond \varphi$ for any $\varphi \in \mathcal{L}$. Then, from $M, w \not\models \|\sim_{\text{Ags}}\|$ it follows there exists acceptable actions $a_1 \in A_1, \dots, a_n \in A_n$, such that $(a_1, \dots, a_n)$ is unacceptable. Then, the set of deontically ideal outcomes in G is not interchangeable. From this, it follows that there exists at least one non-empty critical set $X \subseteq \text{Ags}$.

Let $\mathcal{C} = X_1 \cup \dots \cup X_n$ be a union of all critical sets in G . By construction of $[\mathcal{X}]$, every agent in $\mathcal{C}$ semi-privately discloses her strategy to all members of her critical set, *including herself*. Then, for every $j \in \mathcal{C}$, for any $t \in W$, $v \sim_j^! t$ iff $\sigma(t)(j) = w_j$. In other words, after the sequence of announcements, every member of $\mathcal{C}$ considers a single action of herself possible: the one that she does in w (and given that $M, w \models \delta$, those actions are acceptable). Then, trivially $M_!, v \models (\bigwedge_{j \in \mathcal{C}} \hat{K}_j(a_j \wedge p)) \rightarrow \Diamond(\bigwedge_{j \in \mathcal{C}} a_j \wedge \delta)$. But since $M_!, v \not\models \|\sim_{\text{Ags}}\|$, there exists some action $b_j \in A_j$ for every $j \in \text{Ags} \setminus \mathcal{C}$, such that $M_!, v \models \bigwedge_{j \in \text{Ags} \setminus \mathcal{C}} \hat{K}_j(b_j \wedge \delta) \wedge \neg \Diamond(\bigwedge_{j \in \text{Ags} \setminus \mathcal{C}} b_j \wedge \bigwedge_{k \in \mathcal{C}} w_k \wedge \delta)$. In other words, there should exist some actions of members of $\text{Ags} \setminus \mathcal{C}$, such that the members consider that actions of themselves acceptable, but the aggregation of these actions with $\sigma(w)(\mathcal{C})$ is unacceptable. Then, there exists an acceptable joint action of $\mathcal{C}$, i.e. $\sigma(w)(\mathcal{C})$, and an acceptable joint action of $\text{Ags} \setminus \mathcal{C}$, i.e. $\prod_{j \in \text{Ags} \setminus \mathcal{C}} \{b_j\}$, such that the composition of these two joint actions is unacceptable. Then, $M \not\models \mathcal{C} \|\sim_{\text{Ags}}\| \{\mathcal{C}\}$. Then, by Lemma 1, there should exist a critical partition C_1, C_2 , such that $C_1 \subseteq \mathcal{C}$ and $C_2 \subseteq \text{Ags} \setminus \mathcal{C}$. Obviously, $C_1 \neq \emptyset$ and $C_2 \neq \emptyset$. But by our assumption, $\text{Ags} \setminus \mathcal{C}$ is a set of agents that are not members of any critical set, i.e. for any critical set C , $C \cap (\text{Ags} \setminus \mathcal{C}) = \emptyset$. Contradiction. Hence, the only possibility left is that $\text{Ags} = \mathcal{C}$. But then every agent should have disclosed her own strategy to herself, i.e. for every agent there is a single strategy that she considers possible: her acceptable strategy at w . Then, given that $M_!, w \models \delta$, trivially $M_!, v \models \bigwedge_{j \in \text{Ags}} \hat{K}_j(w_j \wedge \delta) \rightarrow \Diamond(\bigwedge_{j \in \text{Ags}} w_j \wedge \delta)$. But it contradicts our initial assumption that $M_!, v \not\models \|\sim_{\text{Ags}}\|$. Hence, $M_!, v \models \|\sim_{\text{Ags}}\|$. Since $v \in W$ was arbitrarily chosen, $M_! \models \|\sim_{\text{Ags}}\|$. Then, since $\sim_{\text{Ags}}^!(w) \subseteq W$ for any $w \in W$, $M_! \models C(\|\sim_{\text{Ags}}\|)$, i.e. $M \models [\mathcal{X}]C(\|\sim_{\text{Ags}}\|)$.

¹³ These assumptions will not lead to loss of generality, since the construction of the sequence of announcements we are about to present can be easily generalized. Our assumption that there are non-empty critical sets is harmless, since if the only critical set is $\emptyset$, then the set of deontically ideal outcomes is already interchangeable, and it is commonly known, hence, no communication is required. Our assumption that every agent has just two strategies is harmless, since, as the reader will see further in the proof, we will construct the needed sequence of announcements from disclosures of agents' strategies. For generalization, we should replace any announcement of an agent's strategy with its semi-private disclosure.

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